



NAME: INDEX NO:
 SCHOOL: CANDIDATE SIGN:
 DATE:

451/1
 COMPUTER STUDIES
 PAPER 1
 (Theory)
 TIME: 2 ½ HOURS

Kenya Certificate of Secondary Education (KCSE)

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INSTRUCTIONS TO CANDIDATE:

- Write your **name**, **index number** and **school** in the spaces provided.
- This paper consist of **two** sections A and B.
- Answer **ALL** the questions in section **A**.
- Answer question **16** (compulsory) and any other **three** questions from section **B**.
- All answer **MUST** be written in the spaces provided in this question paper.

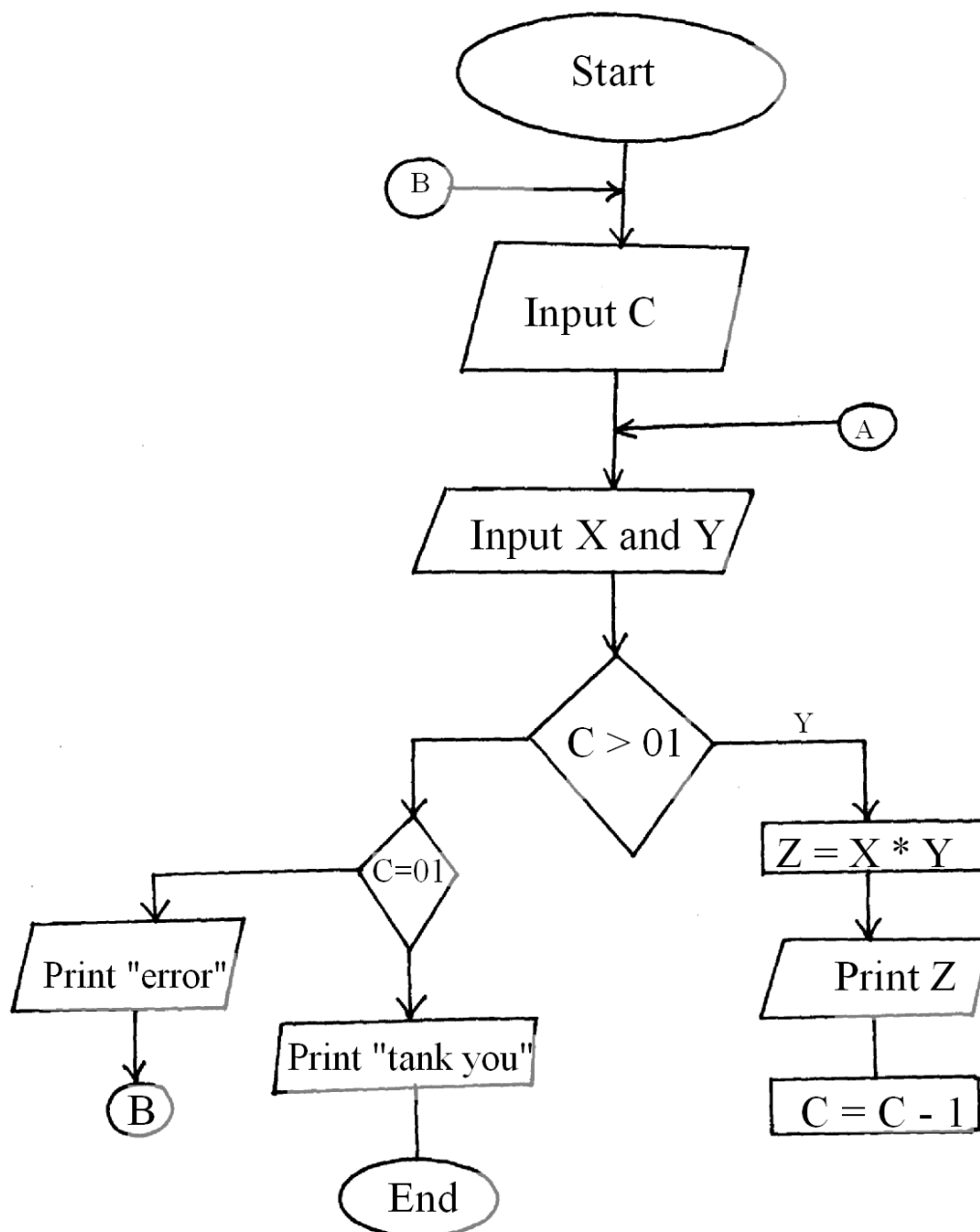
SECTION	QUESTION	ACTUAL SCORE
A	1 – 15	
B	16	
	17	
	18	
	19	
	20	
TOTAL SCORE		

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SECTION B

Answer question 16 and any other three questions in this section.

16. a) State two types of selection controls used in high level programming language. (2 mks)
- b) Give two characteristics of a good program. (2 mks)
- c) Use the flow chart below to answer the questions that follow:





- i) Write a pseudocode for the above flow chart. (6 mks)
- ii) Write the expected output if $c = 2$, $x = 2$ and $y = 3$
- iii) Modify the flow chart so that the program does not accept any negative inputs. (3 mks)
17. a) State and explain two disadvantages that will come about if a network was to be installed in your school. (4 mks)
- b) Discuss two disadvantages of wireless networks. (4 mks)
- c) Write the following abbreviations in full. (4 mks)
- i) F.T.P
- ii) H.T.T.P
- d) With the aid of a diagram, discuss Hybrid topology. (3 mks)
- e) Discuss one advantage of a client/ server network. (2 mks)
18. a) Explain why a computer is able to display the correct time and date when it has just be switched on. (2 mks)
- b) Discuss two types of special memories found I computer system. (4 mks)
- c) i) Define a Bus with reference to a computer system. (1 mk)
- ii) List to examples of buses. (2 mks)
- d) Distinguish between a power cable and interface cable. (2 mks)
- e) Differentiate between the different types of RAM. (4 mks)
19. a) i) Define a system. (1 mk)
- ii) Explain system entropy. (1 mk)
- b) State three circumstances that can lead to development of information systems. (3 mks)
- c) Distinguish parallel changes over from straight change over as used in system implementation. (2 mks)
- d) Discuss two fact finding methods. (4 mks)
- e) Differentiate an open system from a closed system. (2 mks)



- f) List two responsibilities of a system analyst. (2 mks)
20. a) State two ways of representing a signed number. (2 mks)
- b) Write the following abbreviations in full. (2 mks)
- i) ASCII
- ii) EBCDIC
- c) Convert 110011.0110 to Decimal
- d) Differentiate between absolute value and base value. (2 mks)
- e) Outline two reasons for using binary system in computers. (2 mks)
- f) Perform $111011001000_2 - 11011011001_2$ using ones complement convert your answer to Hexadecimal. (4 mks)